Akshay Patil

Dallas, TX | US Citizen | (469) 594-5077 | apatil433@gatech.edu | linkedin.com/in/akshaypatil08 | randomfalcon124.github.io

EDUCATION

Georgia Institute of Technology, Atlanta, GA **May 2029**
Bachelor of Science in Aerospace Engineering | Sophomore | GPA: 4.00
Current coursework: CS 1371 MATLAB, MATH 2551 Multivariable Calculus, COE 2001 Statics, ME 1670 SolidWorks, AE 1601 Intro to Aerospace Engineering, MSE 2001 Materials

School of Science and Engineering at Townview, Dallas, TX **Aug 2022 – May 2026**
3.97 Unweighted GPA | Rank 2 | 1600 SAT | National Merit Scholar | FIRST Alum

EXPERIENCE

YC-Backed Stealth Startup | Hardware Intern **Aug 2026 – Present**
- Building the world's fastest RC interceptor aircraft
- Designed the nose cone and avionics bay, replacing adhesive mounting so the airframe can be built at scale
- Designed 3D printed rib alignment jigs and built two RC test aircraft on them, drilling, finishing, and epoxy bonding carbon fiber structure
- Responsible engineer for the next two programs: a subscale supersonic demonstrator and a fully 3D printed jet interceptor

Hardpoint Robotics | Co-Founder **Jun 2026 – Present**
- Break-even startup building NDAA-compliant development UAVs for customers barred from foreign hardware
- Modeled a 500 mm, 4 kg max takeoff weight quadcopter in CAD with 3D printed CF-Nylon parts, sized for 12.7 min endurance with a 410 g Jetson and stereo camera payload
- Working with North American vendors to manufacture carbon fiber structures and source compliant components, negotiating pricing and delivery timelines

SWYFT Robotics | Engineering Design Intern **May 2025 – May 2026**
- SWYFT sells competition drivetrain and structural parts to FIRST Tech Challenge and FIRST Robotics Competition teams
- Designed sprockets, gears, and pulleys for competition loads, generating $10,000 in revenue

PROJECTS

Autonomous House-Mapping Robot | Personal Project **2025 – Present**
- Jetson Orin Nano robot that explores a house with no operator and counts windows, using an MPPI controller, RTAB-Map visual SLAM, and frontier exploration I wrote
- No lidar: obstacles come from monocular depth classified against a fitted ground plane, windows from three open-vocabulary detectors filtered by a vision-language model

Pickleball Ball Boy UAV | Personal Project **May 2025 – Present**
- Benchmarked five ball detection and four localization methods; AprilTag localization was the most accurate by a wide margin
- Split the drag model into horizontal and vertical coefficients, bringing landing prediction inside a 20 cm zone

RESEARCH AND CLUBS

Hypersonics VIP, Georgia Tech | Data and Communications Team **Aug 2026 – Present**
- Building a sonic fatigue test facility, where acoustic loading is used to find how structures fail under sustained high-intensity sound
- Designing the sensor package that measures pressure, temperature, and airflow inside the facility

FIRST Tech Challenge Team 7172 (Top 0.1% Worldwide) | Captain and Lead Designer **Sep 2021 – May 2026**
- Lead designer on four robots; 2026 FIRST World Championship Finalist Alliance, top 8 of roughly 7,000 teams
- Designed mechanisms using linkages, counter-springing, pulley systems, and geartrains
- Used industrial arm kinematics to design and build a 6-DoF arm robot that expands from an 18 in cube to 13 ft of reach
- 24 competition wins (2021 to 2026), plus Inspire, Think, Connect, Innovate, Design, Motivate, and Control awards

SKILLS

Robotics: ROS 2 (Nav2), RTAB-Map visual SLAM, MPPI control, AprilTag localization
Programming: Python (NumPy, Pandas, TensorFlow), MATLAB, Java, Git
Hardware: NVIDIA Jetson, Raspberry Pi, Arduino, REV Control Hub
CAD: SolidWorks, Fusion 360, Onshape, Autodesk Inventor, 3D printing
Certifications: FAA Part 107 Remote Pilot, Autodesk Certified User (Inventor)